

GAIA Documentation

Complete Reconstruction

Reference model, Sprint 1, Sprint 2, visual recovery and replacement guide

Version 1.0 | 2 August 2026

Executive summary

This report accompanies the Markdown repository. It reconstructs missing Sprint 1 and Sprint 2 artefacts without presenting them as verbatim recovery. Current reference artefacts retain their stable role; Sprint documents preserve research, critique and unresolved questions.

Package design

Source of truth is Markdown plus Mermaid. PNG renders are included for Word/PDF compatibility. A manifest, checksums, source register and replace instructions are included for complete replacement and version control.

Core identity

GAIA is a local-first Personal AI Operating System composed conceptually of bounded digital collaborators. Human control, replaceability, intentional memory, explicit capabilities, observable behaviour and sustainable complexity constrain architecture.

Sprint 1

Framework research, manifesto research, reuse analysis, architectural critique and architecture discussion guide.

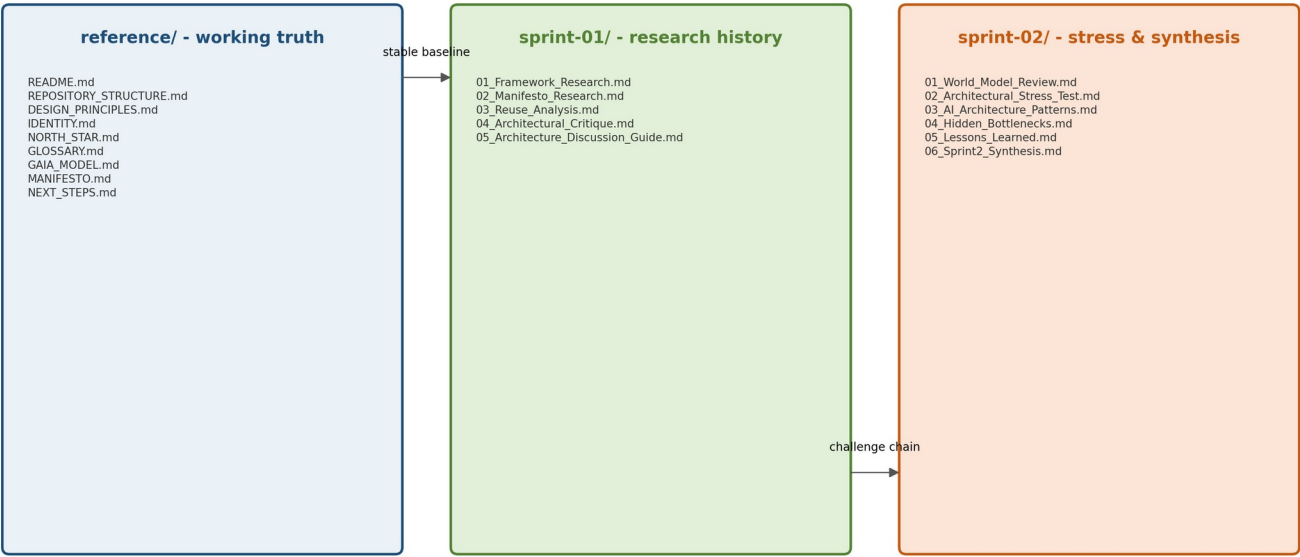
Sprint 2

World-model review, architectural stress test, AI architecture patterns, hidden bottlenecks, lessons learned and synthesis.

Important boundary

The reconstructed Sprint files preserve intent and challenge material from surviving sources and conversation history. They are not claimed as byte-identical copies of lost ZIP contents.

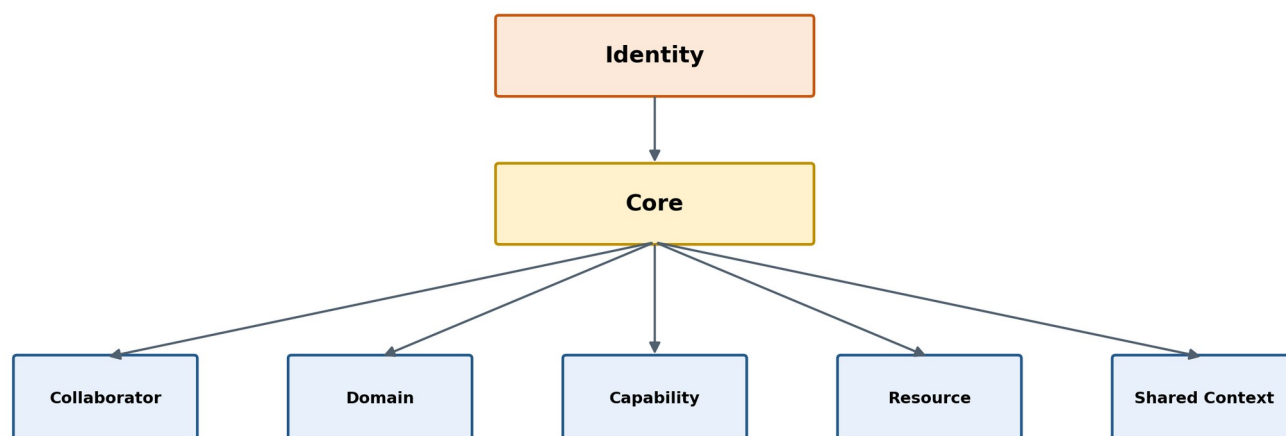
Repository structure and document roles



Sprint material informs future ADRs; it does not overwrite current working truth.

Figure: Repository structure and document roles.

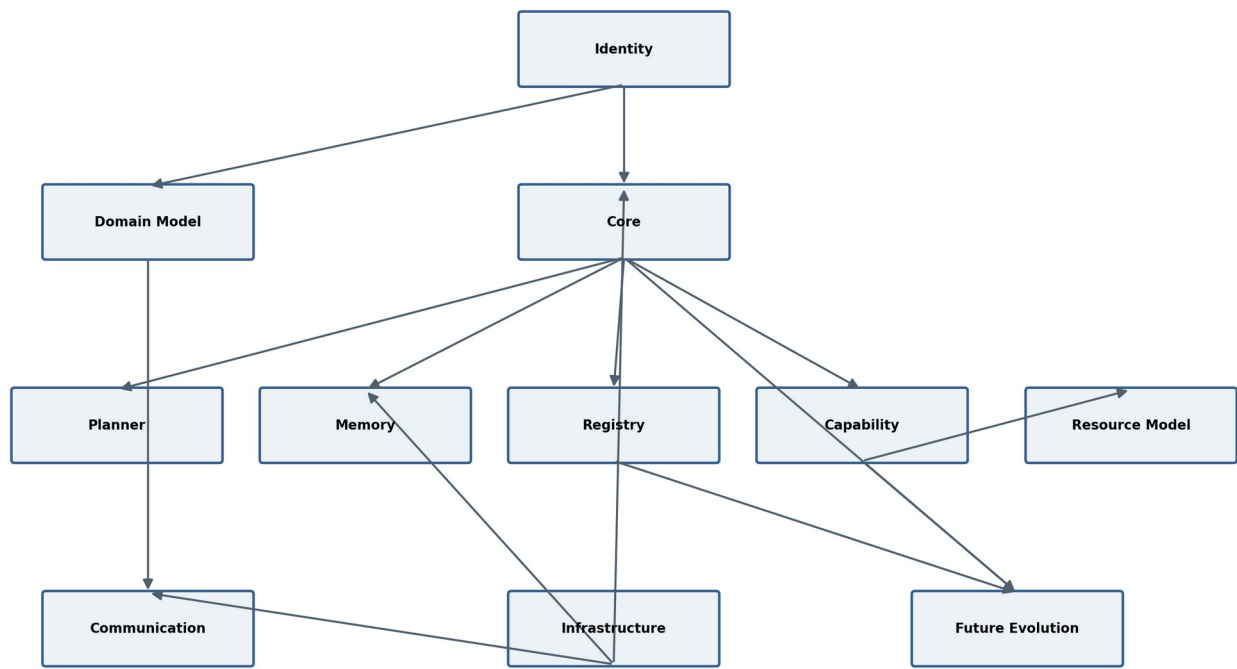
Canonical GAIA conceptual model



Modello concettuale ufficiale: stabile rispetto a framework, runtime e integrazioni

Figure: Canonical GAIA conceptual model.

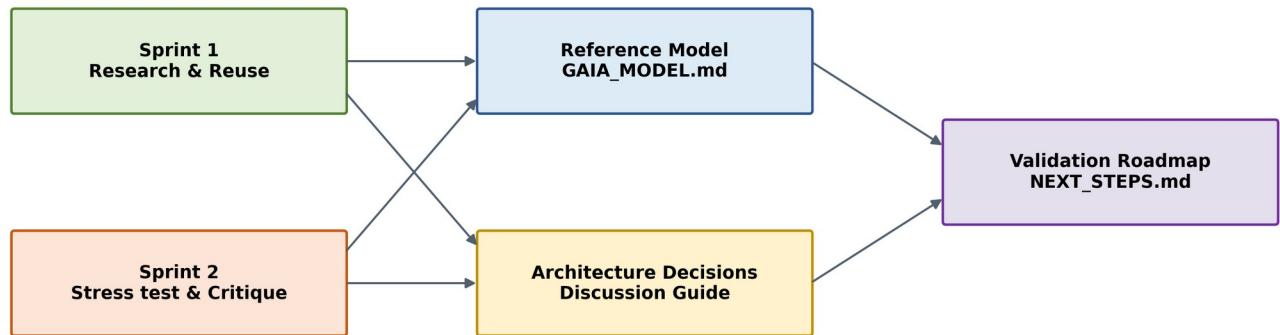
Architecture decision dependency map



Dipendenze decisionali, non flussi runtime

Figure: Architecture decision dependency map.

Sprint-to-reference traceability



Sprint outputs must remain as challenge material; reference documents capture only current working truth.

Figure: Sprint-to-reference traceability.

Maturity roadmap

GAIA maturity path: progress by uncertainty reduction, not feature accumulation



Current research/reference baseline informs later prototype and first-domain validation.

Figure: Maturity roadmap.

Document inventory

reference/

- DESIGN_PRINCIPLES.md
- GAIA_MODEL.md
- GLOSSARY.md
- IDENTITY.md
- MANIFESTO.md
- NEXT_STEPS.md
- NORTH_STAR.md
- REPOSITORY_STRUCTURE.md

sprint-01/

- 01_Framework_Research.md
- 02_Manifesto_Research.md
- 03_Reuse_Analysis.md
- 04_Architectural_Critique.md
- 05_Architecture_Discussion_Guide.md
- README.md

sprint-02/

- 01_World_Model_Review.md
- 02_Architectural_Stress_Test.md
- 03_AI_Architecture_Patterns.md
- 04_Hidden_Bottlenecks.md
- 05_Lessons_Learned.md
- 06_Sprint2_Synthesis.md
- README.md

adr/

- ADR_CANDIDATES.md
- ADR_TEMPLATE.md

incubator/

- IDEAS.md

reports/

- RECONSTRUCTION_REPORT.md
- REPLACE_INSTRUCTIONS.md
- SOURCE_REGISTER.md

prompts/

- NEXT_RESEARCH_SESSION.md

Replacement guidance

Extract the ZIP outside the repository, inspect the manifest, keep a backup of the current folder, then replace the documentation tree. Preserve Markdown and diagrams together. Review the Git diff, scan for secrets and commit explicit paths on a documentation branch.